

DEM generation over the Stromboli summit area using the open source software MicMac : a comparison with Lidar

CÉCILE MÉZON ^{*1,2}, RAPHAEL ANTOINE², CYRILLE FAUCHARD², TULLIO RICCI³,
ADRIEN NORMIER⁴, MARIA MARSELLA⁵, SYLVIA SCIFONTI⁵, and ANTHONY FINIZOLA¹

¹Laboratoire Géosciences Réunion, UR, IPGP, UMR7154, Saint Denis, La Réunion, France

²Centre d'Etude et d'Expertise sur les Risques, l'Environnement, la Mobilité et l'Aménagement (Cerema),
Laboratoire de Rouen, Groupe Sciences de la Terre, CS-90245, 76121, Grand Quevilly, France

³Istituto Nazionale di Geofisica e Vulcanologia, Sezione di Roma I, Rome, Italy

⁴O.R.A. (Observer Relever Analyser) association — "Wings for science" project, Maison des associations,
Paris, France

⁵Dipartimento di Idraulica, Trasporti e Strade, Università I Roma "La sapienza", Rome, Italy

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Abstract

The free open source photogrammetric software called MicMac is tested here as cost-effective tool to generate Digital Elevation Model (DEM) in complex relief areas. A photogrammetric DEM of the Stromboli volcano summit area was built using 39 pictures taken with a Ultralight plane.

To determine the abilities of MicMac in terms of resolution and precision, this model is compared to a Light Detection And Ranging (LiDAR) DEM. The DEMs comparison shows that this photogrammetric approach provides higher resolution (up to 100 times) and a good precision (95.7% within 1 m from the LiDAR).

The lighting of the scene influences the resolution of the DEM, but seems not critical for the accuracy of the DEM. Edges effects are pointed out around areas hidden from the aircraft trajectory (especially because of sharp changes in slope). The DEM presents low accuracy in these areas, but not necessarily a bad resolution.

I. INTRODUCTION

All scientific research dealing with the earth's surface require topographic informations such as elevation, slope and roughness. Photogrammetry, as a powerful method that generates depth maps by means of differently oriented images, allows the calculation of coordinates in a 3D space. This method is here applied to DEM generation, on an area of complex relief. In the last decade, the development of low

cost, high quality digital camera and powerful computers have led to a significant improvement of the photogrammetric method for the reconstruction of 3D structures. Since it is evident that the knowledge of the abilities of the method is necessary to its usage, the present study was undertaken for the purpose of improving the comprehension of the parameters influencing his quality, in terms of resolution and accuracy.

For the most part, DEMs have vertical accura-

*Corresponding author

cies ranging from 1 to 10 m, depending on the methods used to acquire the data [Bolstad & Stowe, 1994; Hodgson *et al.*, 2003]. Vertical accuracies of 15 cm to 1 m could be obtained with LiDAR [Gomes Pereira & Wicherson, 1999; Hodgson *et al.*, 2003; Reutebuch *et al.*, 2003]. Generally, the accuracy of these data increases with the density of initial point [Gong *et al.*, 2000].

Photogrammetry has been compared to others technique, such as Shuttle Radar Topography Mission (SRTM), or interferometric synthetic aperture radar (InSAR) technique for DEM generation [Sefercik, 2007]. This particular study highlight that photogrammetry provides better accuracy for open and flat terrain and for forest areas than the others techniques tested. [Gong *et al.*, 2000] test various factors on the accuracy of the DEMs, such as the sampling interval, inclusion or not of feature points, relief...etc and noticed that generally speaking, the accuracy of DEMs decreases with an increase in relief, even if it could not be the case.

This article propose a comparison of a photogrammetric DEM to the widely used LiDAR on an area of complicated relief, LiDAR acting as the reference method. This kind of comparison is rare, because of the cost of the LiDAR acquisition, and exceptional considering the four months between the two acquisitions. After the results of the comparison, a discussion is proposed to reveal some critical parameters, on such complex topography environment, responsible of modifications of the quality (accuracy and resolution) of the photogrammetric model.

II. DATA ACQUISITION

II.1 Area of interest

The island of Stromboli is the emergent portion of a submarine stratovolcano which rises from the bottom of the Tyrrhenian Sea to 924 m above sea level. Stromboli is the northernmost island of the Aeolian volcanic arc and the studied area is located in the upper part of the edifice (Figure 1a).

Stromboli volcano presents a cone shape

with steep slopes. The investigated zone located near the active vents, is vegetation free and exhibits some sharp changes in slopes. This area ($\approx 5.10^5 \text{m}^2$) could be caricatured as two flanks (slopes about 35°), on both sides of a crest, extending in the NE-SW direction. This crest is actually a structural limit, delineating the upper rim of the Rina Grande sector collapse (Figure 2a). Another structural limit, located Southward from the crest, is the Neo-Stromboli crater (Figure 2a), which represents a lithological boundary. Massive lava flows, characterized by red patina, are located outside (Eastern side) the crater boundary (Figure 2a). These red outcrops, displaying sweeping change in slopes, belong to the Vancori period. The refilling crater area (Western side), also affected by few sector collapses Northwestward, is made of ash and lapilli from the ongoing activity [Francalanci *et al.*, 2013]. The current activity of Stromboli consists of intermittent mild explosions (every 15-20 minutes) and ejects scoriaceous bombs, lapilli, and ash [Ripepe *et al.*, 2001; Chouet *et al.*, 2003].

II.2 Data acquisition

II.2.1 Photographs acquisition

Pictures were taken in the framework of the non-profit scientific flights operated by the association "Wings For Science" on the 27th January 2012 between 11 h 30 and 12 h, from an ultra-light gyroplane with a hand-held camera (Canon EOS 5D, Mark II). The flight crosses the studied area from the South to the North of the volcanic edifice, at an altitude of about 1150 meters a.s.l., by the East flank (Figure 1b).

Unlike other aerial systems, gyroplane offers manoeuvrability at low speed, despite any turbulence. Moreover, modifications of the mission planning can be immediately operated for an efficient observation.

Photographs taken during an explosion or unfavourable light conditions were removed before photogrammetric processing. Finally, 39 pictures with an average pixel area of 5 cm^2 were selected (focal length: 70 mm, number of pixels: 5616×3744 , sensor size: $23.9 \text{ mm} \times 35.8$

mm, exposure time comprised between 1/1000 s and 1/640 s). As it is shown in the Figure 1b, there is a significant overlap between photographs (at least 60%).

Finally, more photographs were chosen to map the North flank, because of the unfavourable lighting conditions (see Figure 2a) caused by the shadow of the crest at mid-day (North flank is visible on 33 photographs whereas the South flank is visible on 19 pictures only).

II.2.2 Airborne Laser Scanning (ALS) data acquisition

ALS survey was performed on 5th, 6th and 10th May 2012 using an Optech Gemini ALTM sensor based on a fully automated continuous multi-pulse technology that provided very dense and accurate 3D points. The nominal accuracy of the coordinates is ranging between 0.05 and 0.30 m in the horizontal and vertical direction depending on the selected operational parameters. The survey, performed at about 1500 m a.s.l. and adopting a nominal field of view of 50°, collected data at a density up to 12 points/m² along 15 overlapped strips covering the entire island. The ALS sensor was coupled to an high resolution digital camera ADS80-SH82 that acquired images adopted for rendering the surface model and to extract orthophoto maps.

III. DATA PROCESSING

III.1 Photogrammetric treatment

Apero-MicMac (*Aérotiangulation Photogramétrique Expérimentale Relativement Opérationnelle - Multi-Images Correspondances, Méthodes Automatiques de Corrélation*) is a free, open-source and multi-view stereo software [De-seilligny & Clery, 2011](more informations are available at [http : //www.micmac.ign.fr/](http://www.micmac.ign.fr/)) which generates 3D models out of pictures taken from arbitrary positions. Unlike other commercial and open-source photogrammetric softwares, Apero-MicMac is less user-friendly,

but its performances are built on strong mathematical development and regular updates, tested in various applications [Fauchard *et al.*, 2013, Bretar *et al.*, 2013]. The code is also highly adaptable to various acquisition configurations. Three main steps lead to DEM generation: (i) scale-invariant feature transform algorithm (SIFT)[Lowe, 2004] is performed to detect tie points on pairs of photographs; (ii) pictures are oriented one after the other, relatively to a master image and according to the number and distribution of previously detected tie points. At this stage, GPS measurements or ground control points may be taken into account; (iii) an accurate and dense depth map is built with the calculated orientations and then converted into a 3D point cloud by means of an energy minimisation algorithm solved by a multi-scale dynamic programming.

The only camera's parameters needed by MicMac are the focal length and the sensor size.

III.2 LiDAR treatment

The DEM quality assessment was performed using a validation dataset compounded by previous ALS, in order to evaluate the repeatability on stable areas (man-made structures and rocks), and on a number of GPS control points. This analyses provided an estimate for the average horizontal and vertical accuracy of 0.11 and 0.15 m, respectively.

IV. RESULTS AND DISCUSSION

The 3D point cloud generated by MicMac (Figure 2a) consists of about 9.10^6 points and was georeferenced with three noticeable ground control points. Despite the small amount of GCP for the georeferencing process, the operation was realized on noteworthy points (shelters, pedestrian pathway crossing) (Figure 3a). Consequently the order of magnitude of the error made by the georeferencement is in the range of few pixels (pixel size: 2.3 cm). All point cloud characterization were made using Cloud CompareTM, an open-source 3D point cloud and mesh processing software. To calcu-

late the DEM point density (in pts/m²), Cloud CompareTM counts for each point belonging to the cloud, the number of surrounding points included in a sphere whose disc area is 1 m². The point density of photogrammetric DEM is shown in the Figure 2b.

The density map presents a dichotomy between the North flank and the South flank of the crest (Rina Grande sector collapse). The South flank exhibits a wide range of densities, ranging from 0 (blanked areas) to 651 points/m². Blanked areas are exactly located around the reliefs of Neo-Stromboli outcrops (O1, O2 and O3 in the Figure 2). They correspond to hidden surfaces from the aircraft trajectory.

On the other side, the North flank displays a narrower range of densities, ranging from approximately 20 to 100 points/m². The difference of ranges between North and South slopes could depend on (1) the aircraft trajectory, gradually moving away from the studied area producing therefore differences in the pixel size of photographs, or on (2) the shadow generated by the crest on the North flank. The first option seems unlikely to be of primary order, considering the fact that the photographs were taken in burst mode (3 im/s). The second one is more realistic pointing out that the lighting of the scene could be a critical parameter for MicMac processing, making tie points more difficult to find.

In contrast, the density map of the LiDAR DEM (Figure 3b) is very homogeneous, with densities ranging from 1 to 8 points/m². The density map of the LiDAR DEM is directly linked to the slope map (Figure 3a): lowest slopes give higher resolution especially on the crest (Rina Grande sector collapse). This correlation is due to the acquisition protocol, the flight altitude being constant, and so independent from the soil slopes.

As clearly shown in Figure 2b and 3b, the resolution of the photogrammetric DEM can be a hundred times higher than the one of the LiDAR. In turn, the survey was unable

to image the entire scene, and blanked areas appear on the photogrammetric DEM, which is not the case with the LiDAR protocol.

To deepen this analysis, a two-stage comparison study was performed by considering the LiDAR DEM as the reference data. First, the 3.10⁵ points of LiDAR DEM were meshed according to a Delaunay triangulation tool [Kazhdan et Hoppe, 2013] to be compared with the 9.10⁶ points of photogrammetric DEM. We first calculated the mesh to cloud distances to test the LiDAR point cloud triangulation efficiency (more information on distance computation [http : //www.geometrictools.com/Documentation/DistancePoint3Triangle3.pdf](http://www.geometrictools.com/Documentation/DistancePoint3Triangle3.pdf)). The computation of the signed distances between the LiDAR mesh and the LiDAR point cloud was defined as the length between each point of the LiDAR point cloud to the nearest surface of the LiDAR mesh. The distribution of the signed distances is centered on 0.01 cm and presents a standard deviation of 6.6 cm, so an error lower than the LiDAR precision.

Second, the signed distances between the meshed LiDAR DEM and the point cloud photogrammetric DEM were computed. The signed distances follow a normal distribution centered on 0.06 m with a standard deviation of 0.45 m (Figure 4a), meaning that more than 95% of the photogrammetric DEM points are closer than one meter from reference points. This calculation validates the consistency of the photogrammetric method as an accurate and powerful tool for DEM generation.

Nevertheless, there are in the Figure 4a, two distinct kinds of anomalies. The first one (A1) is spread out on the Northwestern flank, just near the crest and persists on the Eastern side, giving to this anomaly the shape of a checkerboard. The other ones, presenting higher amplitudes, are located on the Southern flank, around Neo-Stromboli outcrops (O1, O2, O3 in the Figure 4a).

The A1 anomaly is characterized by signed

distances ranging from 0.3 m to 1.7 m and an important spatial extension. The noticeable appearance of a grid pattern, centered on the Eastern GCP, is known to be due to the approximation done by Aperio-MicMac during the georeferencing procedure (GCPBascule). The UTM kilometric grid used for the georeferencing displays large numbers (here around 4294000 in the Y direction) whereas GCPBascule use a 32 bits floating point to encode the coordinates. For coordinates around millions, the least difference that the code can memorise is 0.5 m. Moreover, winter winds blowing in the South-North direction may add to the amplitude of this anomaly. Those winds can remove ashes from the South flank to the North flank. This reshuffle could explain up to 50 cm of accumulation of volcanic ash on the Northern flank, close to the crest (T.Ricci, personal communication). These ashes, as any fine particle located on a slope, could easily be removed with rainfalls, explaining difference in elevation between winter and spring.

The anomalies O1, O2 and O3, observed on the Southern flank are all located around the Neo-Stromboli outcrops. The distance between DEMs could reach 3 m, and sometimes turns from one pixel to the other from 3 m to -3 m. The areas characterized by such distances are mostly located on areas of steep slopes (until 80°). These anomalies might be due to the resolution difference between the two DEMs, and the approximation made by the LiDAR mesh. In fact, when the surface is interpolated from the LiDAR point cloud by three points, there are significantly more points on the photogrammetric DEM to represent the surface. When the slope reaches such values ($60-80^\circ$), a minor difference in a horizontal direction, leads to a significant difference in the vertical direction, causing this kind of anomaly, positive and negative side by side.

Furthermore, these anomalies are exactly located around blanked areas. There seems to be an edge effect around the blanked zones of the photogrammetric DEM (Figure 2). But regarding to the density map of the photogrammetric DEM (Figure 2b), all blanked areas are

surrounded by bad resolution ones. This information means that all photogrammetric poorly resolved areas are not equivalent. In fact, the Northern flank exhibits bad resolution, but nevertheless, presents good correlation with the LiDAR DEM. On the other side, each blanked area on the photogrammetric DEM are fringed by areas of bad resolution which correspond to the less precise areas.

This information also explains why there is no correlation between the point density of the photogrammetric DEM, and the map of distances between the two DEMs (Figure 4b). This non correlation is of big interest and means that even if MicMac only finds few tie points, they will not necessarily be misplaced. This proves the effectiveness of MicMac and our difficulty to find accuracy indicators of the photogrammetric method.

V. CONCLUSION

This work has focused on a static validation of the photogrammetric method in generating high quality DEMs. The photogrammetric model shows a good correspondence with the LiDAR DEM (95.7% of the points are located less than one meter from the LiDAR DEM). In particular two areas present bad correlations and could be explained by the quality of pictures, their ability to image the scene, or by the steepness of the site. Nevertheless, these kind of inconvenients could have been avoided by a better aircraft trajectory or better lighting conditions.

We can not actually determine if the difference in resolution between the north flank and the south flank is directly due to the lighting conditions or to difference in lighting conditions between both flanks. Anyway, the difference in resolution does not substantiate for an equivalent accuracy. A better aircraft trajectory would have allowed to map the whole scene; and so might prevent blanked areas and inaccurate areas fringing them.

Both methods, LiDAR and photogrammetry presents advantages and inconvenients and

seem to be complementary. To built high resolution DEMs, it would be of major interest to combine the two methods, to ensure the precision of the LiDAR technique, with the resolution of the photogrammetric method [Fabris et al. 2010].

Finally, photogrammetry seems very promising, particularly for his ability to produce ultra densified 3D model. Even if such resolution of the models does not substantiate for an equivalent accuracy, it remains that photogrammetry is a financially viable method to produce high quality DEM especially for low extension inaccessible regions.

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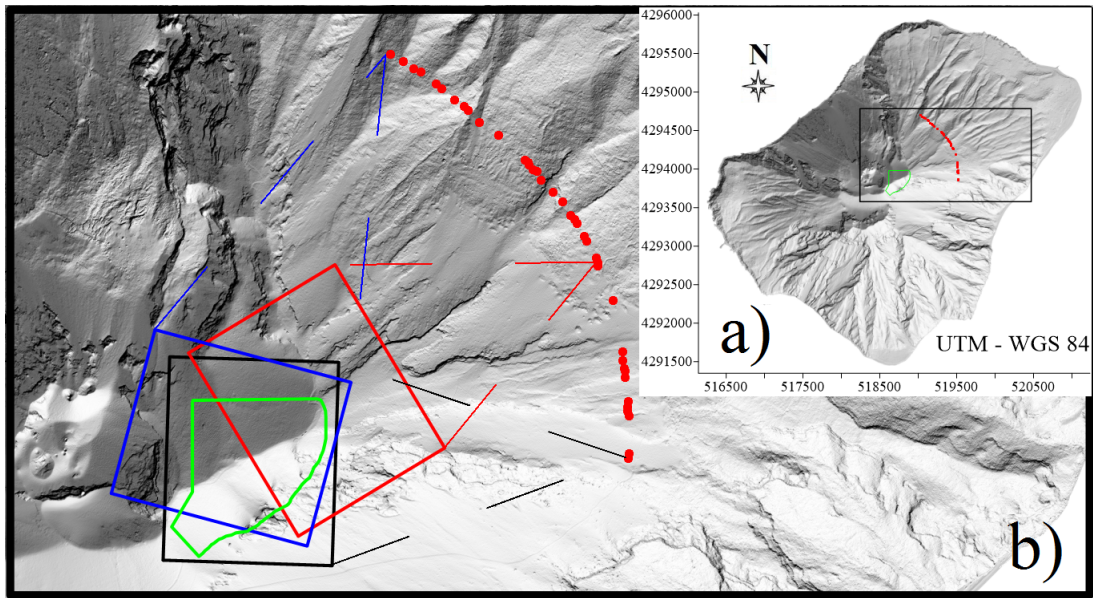


Figure 1: *a) Shaded relief map of Stromboli volcano obtained by LiDAR. The area surrounded in green fits the studied area. Red dots represent camera's position. b) Zoom over the area of interest : colored rectangles represents the soil surface catch on photographs.*

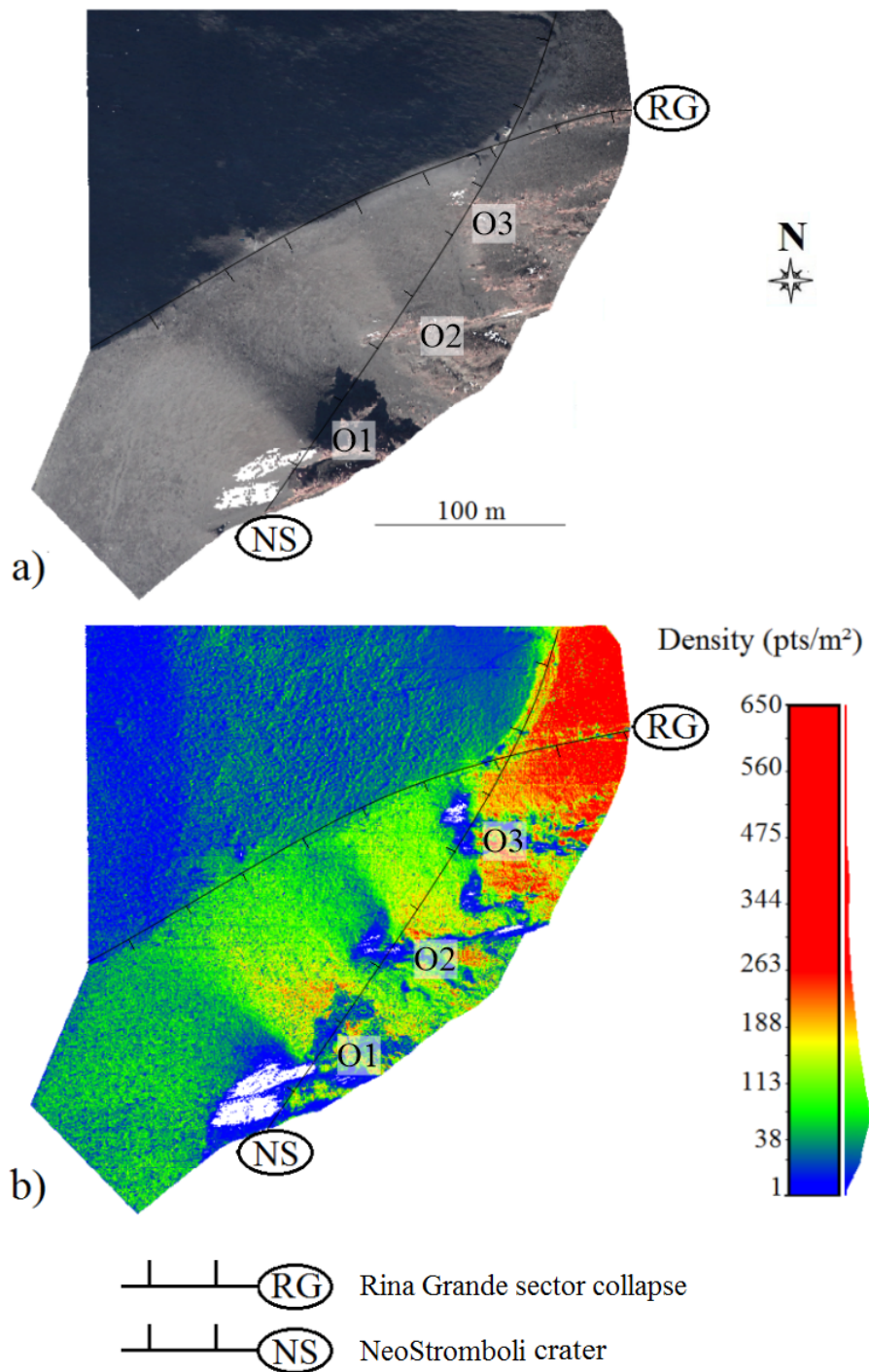


Figure 2: a) DEM point cloud after the complete photogrammetric process. The model is made of more than 9 millions of points over a surface of about 5.10^5 m^2 . b) Point density of the photogrammetric DEM. The two main structural boundaries are drawn (the Rina Grande sector collapse and the NeoStromboli crater). The blanked areas correspond to no data areas.

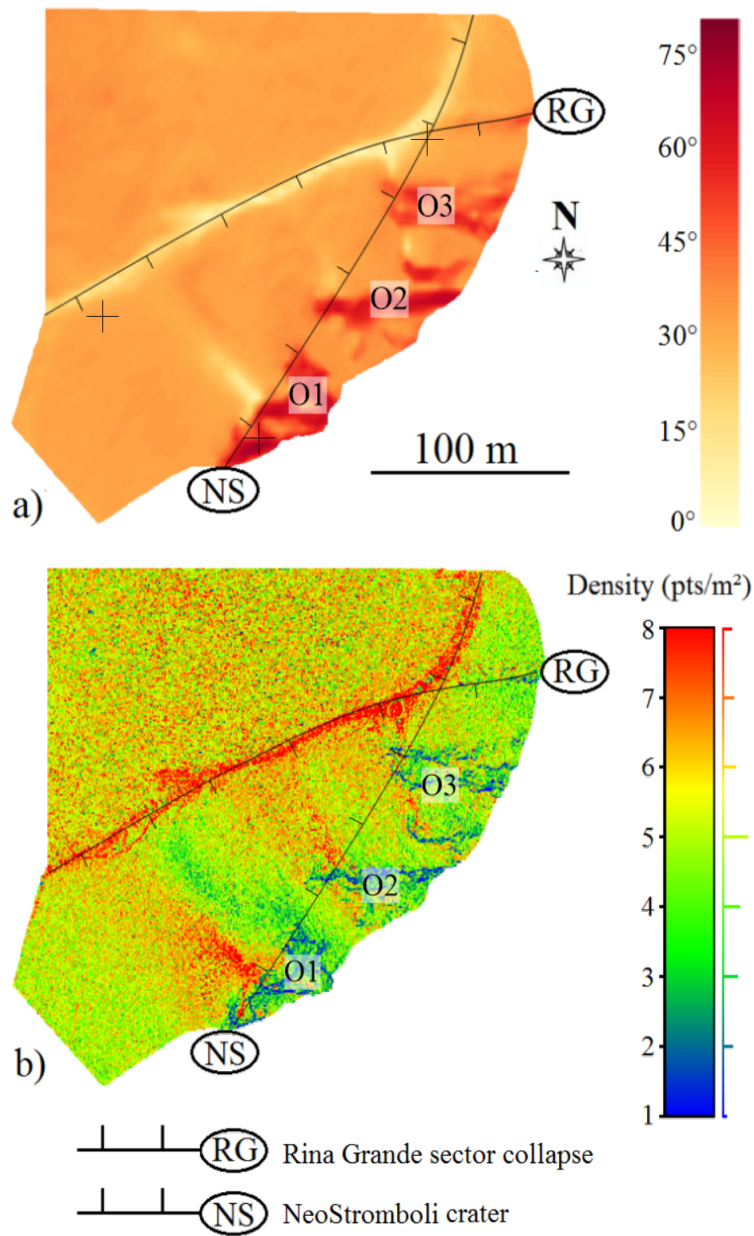


Figure 3: a) Slope map of the area of interest calculated from the LiDAR DEM. Black crosses represents GCP positions.
b) density map of the LiDAR DEM using the same process that the photogrammetric model.

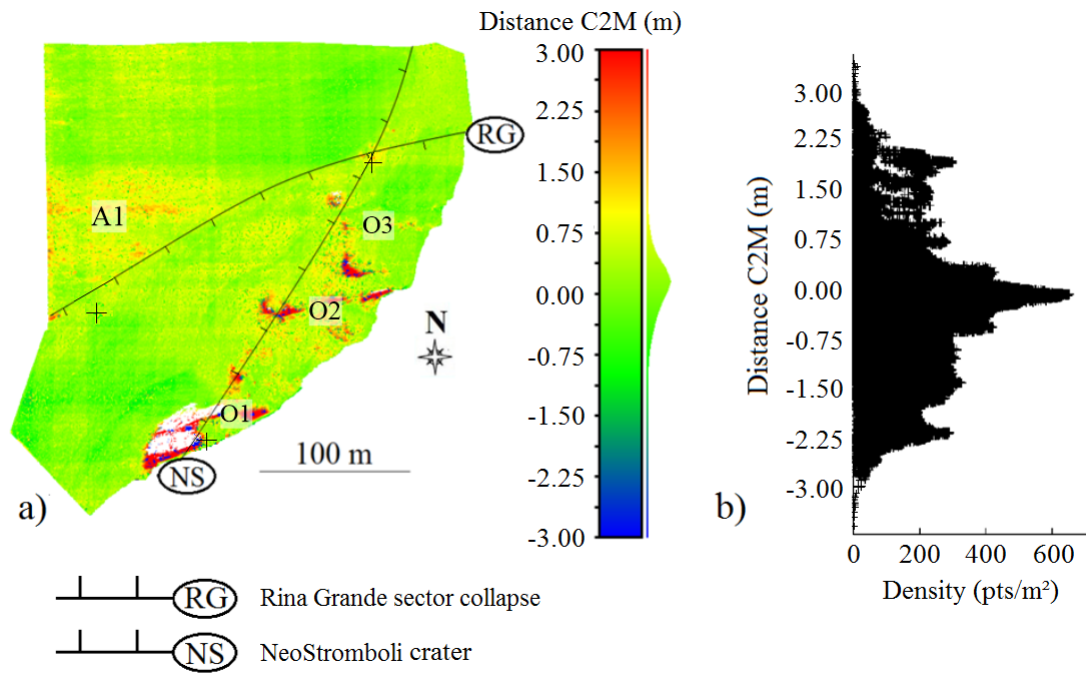


Figure 4: a) The map of signed distances between the meshed LiDAR DEM and the photogrammetric point cloud, LiDAR DEM being the reference mesh. Black crosses represents GCP positions. b) Signed distances between the two DEM vs the photogrammetric point density. Correlation coefficient : 0.063